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Light Gradient Boosting Machine: An efficient soft computing model for estimating daily reference evapotranspiration with local and external meteorological data

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ABSTRACT

Accurate estimation of reference evapotranspiration (ETo) is required in many fields, e.g. irrigation scheduling design, agricultural water management, crop growth modeling and drought assessment. Nevertheless, reliable estimation of ETo is difficult when lack of complete or long-term meteorological data at the target station. This study evaluated the efficiency of a new tree-based soft computing model, Light Gradient Boosting Machine (LightGBM), for estimating daily ET_0 using limited local (target-station) and external (cross-station) meteorological data from 49 weather stations in humid subtropical region of China, including 16 in Jiangxi Province and other 33 in the region. The performance of LightGBM was compared with the tree-based M5 Model Tree (M5Tree) and Random Forests (RF) as well as four empirical models (Hargreaves-Samani, Tabari, Makkink and Trabert). Eight input combinations of daily meteorological data including maximum temperature (T_{max}), minimum temperature (T_{min}), relative humidity (H_r), wind speed at 2 m height (U_2), extraterrestrial solar radiation (R_a) and global solar radiation (R_s) calculated from sunshine duration (n) for the period 2001–2015 were used to test the models. The results showed that LightGBM was superior to M5Tree and RF in local applications under all input combinations during testing, with average root mean square error (RMSE) of 0.08–0.58 mm d^{-1}, 0.11–0.62 mm $d^{-1}$and 0.13–0.60 mm d^{-1}, respectively. M5Tree performed slightly better than RF under input combinations 6–8, whereas RF outperformed M5Tree under the other input combinations. However, all three soft computing models produced much better daily ET_o estimates than the corresponding empirical models with the same input variables. R_s was the most influential meteorological variable for daily ETo estimation in this region, followed by T_{max} and T_{min}, H_r and finally U_2. In external applications, LightGBM also generally performed better than the RF, M5Tree and empirical models. Soft computing models developed with meteorological data from Station 57894, having the most similar climatic characteristics to the other stations, gave satisfactory ETo estimates for the 15 cross stations in Jiangxi Province, even for the other 33 stations across the humid subtropical region of China. LightGBM was proved to be efficient and exhibit good generalization capability in both local and external applications, which was thus recommended as an alternative model for daily ETo estimation.

1. Introduction

Quantitative determination of reference crop evapotranspiration (ETo) is necessary in many fields, e.g. irrigation scheduling design, water resources management, crop growth modeling and drought assessment (Feng et al., 2016, 2017; Wu et al., 2019a). However, direct measurement of ETo is difficult, time-consuming and costly (Irmak et al., 2003). The alternative procedure is to estimate ETo from local readily available meteorological variables. The concept of potential evapotranspiration (ET_p) was originated from the Penman combination based method (Penman, 1948). For the practical reasons, ETo was proposed in the 1970s as an alternative to the term ET_p. Many other empirical models such as temperature-based, radiation-based and mass transfer-based models, have been developed and modified for

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estimating ETo. The FAO-56 Penman Monteith equation (FAO-56 PM) has been recommended as a standard method for estimation of ETo around the world by the Food and Agriculture Organization of the United Nations (FAO). The FAO-56 PM equation requires many climatic parameters, such as net radiation, wind speed, air temperature and relative humidity. However, some of these input variables may not be available for a given location. In developing countries in particular, the quality of data and difficulties in collecting all the required meteorological parameters can pose serious constraints on the application of FAO-56 PM model. For instance, the China Meteorological Administration only runs 130 radiation stations, and the existing methods for estimating solar radiation from weather data are not accurate enough in many parts of China (Fan et al., 2020). In addition, although wind speed prediction has been extensively studied in non-agricultural fields (Yang and Wang, 2018; Talaat et al., 2019), there is insufficient evidence that these methods are suitable for ETo estimation.

When the FAO-56 PM method cannot be implemented for a particular area, the utilization of simplified temperature-based, radiation-based or mass transfer-based methods is a common choice. The core idea of temperature-based empirical models is to reduce the input data as much as possible according to regional climate characteristics on the premise that the impact on prediction accuracy is not very large. Hargreaves and Samani (Hargreaves and Samani, 1985) combined the original ETo equation introduced by Hargreaves (Hargreaves, 1975) and another equation (Hargreaves and Samani, 1982) to estimate solar radiation based on temperature data. Due to its simple structure and high precision in many regions of the world, the application of Hargreaves-Samani model has become increasingly popular among the empirical models (Shahidian et al., 2013; Heydari and Heydari, 2014; Shiri et al., 2015a, 2015b; Almorox and Grieser, 2016; Feng et al., 2017; Paredes et al., 2018). Other researchers have also established empirical models or simplify Penman-type models based on solar radiation data, e.g. Makkink model (Makkink, 1957) and Tabari model (Tabari et al., 2012). Comparison, calibration and modified versions of these radiation-based models have also been largely reported (Rahimikhoob et al., 2012; Tabari et al., Traore et al., 2016Chatzithomas and Alexandris, 2015; Feng et al., 2016, 2017; Shiri, 2017; Tongwane et al., 2017; Antonopoulos and Antonopoulos, 2017; Almorox et al., 2018). The mass transfer method (aerodynamics) is based on the quantitative description of the physical process of water vapor transfer from the evaporating surface to the atmosphere in the eddy motion. These methods obtained satisfactory results under the simulated water surface evaporation (Jensen, 1973; Linacre, 1994; Singh and Xu, 1997), but many studies have shown that such methods are less accurate than radiation-based and temperature-based methods (Jakimavičius et al., 2013; Traore and Guven, 2013; Djaman et al., 2015, 2017; Mehdizadeh et al., 2017a, 2017b; Lu et al., 2018).

Recently, soft computing models are found to show excellent reliability in ETo estimation and forecasting (Gocić et al., 2015). There are a variety of soft computing models based on different theories, e.g. Artificial Neural Networks (ANNs) (Landeras et al., 2008; Yassin et al., 2016; Traore et al., 2016), Gene Expression Programming (GEP) (Yassin et al., 2016; Shiri, 2017; Traore et al., 2017; Mattar, 2018), Support Vector Machines (SVM) (Kisi, 2013; Wen et al., 2015; Kisi and Parmar, 2016), Extreme Learning Machine (ELM) (Abdullah et al., 2015; Feng et al., 2016, 2017; Fan et al., 2018b), Multivariate Adaptive Regression Splines (MARS) (Kisi and Parmar, 2016; Mehdizadeh et al., 2017a,2017b; Wu and Fan, 2019) and Adaptive Neuro-Fuzzy Inference System (ANFIS) (Shiri et al., 2015a, 2015b; Gong et al., 2016; Keshtegar et al., 2018; Tao et al., 2018). Besides these methods, the tree-based machine learning models are gaining in popularity due to their extremely high speed and satisfactory accuracy (Feng et al., 2017; Shiri, 2018; Fan et al., 2018b). Kisi and Parmar (2016) compared the performance of least-squares SVM, MARS and M5 Model Tree (M5Tree) for ETo estimation in two cities of Turkey, and found that M5Tree performed better than others when local input and output data were unavailable. Feng et al. (2017) compared the applicability of Random Forests (RF) and Generalized Regression Neural Network (GRNN) for estimating ETo in Sichuan of China with only temperature data as inputs. They found that both models had better prediction accuracy in local and cross-station applications, while RF was slightly superior to GRNN. The RF model coupled with wavelet method improved the prediction accuracy of ETo when using input variables same as those of mass-transfer empirical models (Shiri, 2018). Due to the best combination of estimation accuracy, model stability and computational efficiency, the tree-based Extreme Gradient Boosting (XGBoost) and Gradient boosting decision tree (GBDT) were recommended for estimation of daily ETo in China (Fan et al., 2018b).

Light Gradient Boosting Machine (LightGBM) is an improved version of gradient learning framework based on decision trees and the idea of “weak” learners. Since being developed by Microsoft in 2017 (Ke et al., 2017), LightGBM has been widely applied in many fields as a result of high prediction accuracy, fast computational speed and excellent capability of minimizing over-fitting issues relative, e.g. web search, miRNAs identification in breast cancer (Wang et al., 2017a), P2P network loan default (Ma et al., 2018), music recommendation (Zhang et al., 2018a, 2018b), acoustic scene classification (Fonseca et al., 2017) and smart grid load forecasting (Zhang et al., 2018a). To the knowledge of the authors, LightGBM has not yet been applied in agricultural or hydrological studies, especially in ETo modeling. Besides, the estimation of ETo for a target station is always difficult when long-term local data are missing or unavailable, especially in China and other developing countries. In this situation, data supplement from neighboring stations are needed for estimation of local ETo (Wang et al., 2017a, 2017b; Karimi et al., 2017). Thus, the objectives of the present study are: 1) to explore the capability of the new LightGBM model for estimating daily ETo with limited local meteorological data in the humid subtropical Jiangxi Province of China; 2) to compare the prediction accuracy of the LightGBM model against two traditional tree-based soft computing models (i.e. M5Tree and RF) along with four commonly used empirical models (Hargreaves-Samani, Tabari, Makkink and Trabert) for estimation of daily ETo; and 3) to further investigate the potential of the proposed empirical and soft computing models for estimating daily ETo using external (cross-station) meteorological data.

2. Materials and methods

2.1. Data collection and input combinations of meteorological variables

The present study was carried out in the humid subtropical region of China, including 16 weather stations in Jiangxi Province of China and other 33 weather stations across the region (Fig. 1). Jiangxi Province was surrounded by mountains on three sides except north, with plains in the north and hills in the center. Station 58506 has the highest elevation, which is located on the top of a mountain. Table 1 sums up the geographical and meteorological information of the 16 stations in Jiangxi Province of China. The corresponding information of other 33 weather stations across the region is given in Table S1. Continuous daily meteorological data during the period 2001–2015 were collected in this study, including daily maximum and minimum temperature (T_{max} and T_{min}), sunshine duration (n), relative humidity (H_r), wind speed at 2 m height (U_2) and extraterrestrial solar radiation (R_a). Global solar radiation (R_s) was calculated from n using the Angstrom-Prescott model according to Fan et al. (2019a). Meteorological data were provided and quality-controlled by the National Meteorological Information Center of China Meteorological Administration. It can be seen from Table 1 that Station 58506 and 58634 experienced relatively lower annual mean T_{max} (16.3 °C and 19.6 °C, respectively) and T_{min} (9.6 °C and 11.7 °C, respectively) compared with those at the other stations (22.4–24.9 °C and 13.1–16.8 °C, respectively). Lower annual mean n (3.4 h d^{-1}) but higher RH (86.4%) was observed at Station 58634

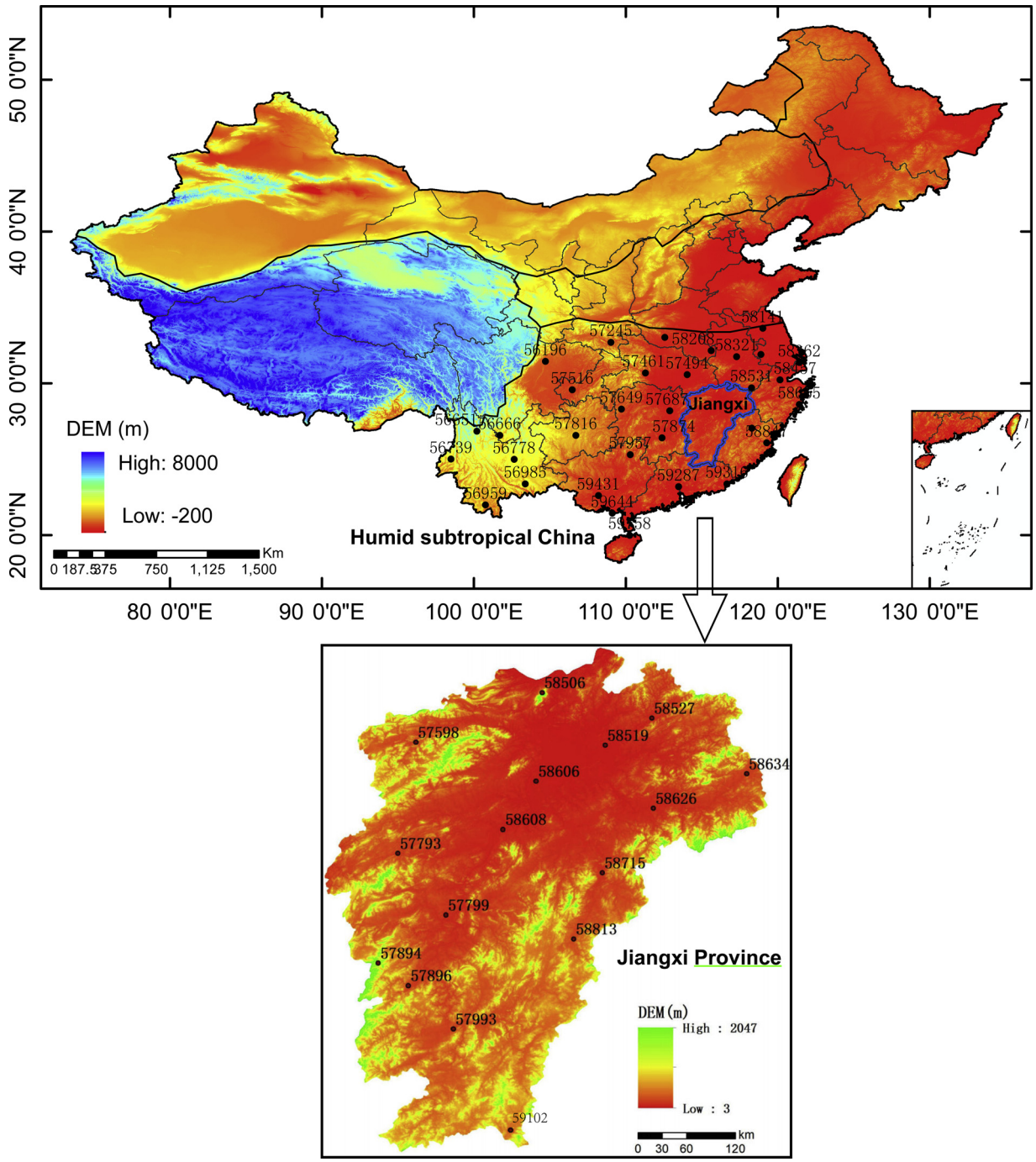


Fig. 1. Geographical locations of the 49 weather stations in the humid subtropical region of China.

relative to those at the other stations (4.0–5.4 h $d^{-1}$and 71.2%–80.8%, respectively). However, Stations 58506 (3.6 m s^{-1}) and 58715 (2.6 m s^{-1}) received relatively higher annual mean U_2 than the other stations (1.0-2.0 m s^{-1}). The lowest annual mean ET_0 (1.8 mm d^{-1}) was obtained at Station 58634 in Jiangxi Province as a result of low n and high RH, while the other 15 stations had comparable annual mean ET_0 (2.2-2.7 mm d^{-1}). A comparison of Tables S1 and 1 indicated that the values of meteorological variables over the other 33 weather stations across the humid subtropical region of China were generally similar to those in Jiangxi Province, but larger differences occurred due to much longer distances among the stations. For instance, larger ETo values (> 3.3 mm d^{-1}) were observed at Stations 56651, 56666, 56959, 56985 and 59758. The observed meteorological data from 2001 to 2015 were divided into two datasets. The first dataset (2001–2010) was used to calibrate/train the four empirical models and the three soft computing models, while the second dataset (2011–2015) was used to test the developed models. It can be seen from Table 1 that the differences between the meteorological values during training and testing at each weather station did not vary greatly. In this study, eight input combinations based on air temperature and other meteorological variables were considered as inputs to the three soft computing models (Table 2).

2.2. FAO-56 Penman-Monteith equation (FAO-56 PM)

According to the FAO-56 PM equation (Allen et al., 1998), ETo (mm d^{-1}) was calculated as:

Table 1
Geographic and meteorological information of the 16 weather stations in the humid subtropical Jiangxi Province of China during 2001–2010 (2011–2015).

ID	Latitude (°N)	Longitude (°E)	Elevation (m)	T_{max} (°C)	T_{min} (°C)	n (h d^{-1})	RH (%)	U_2 (m d^{-1})	ETo (mm d^{-1})
57598	29°02'	114°35'	147.4	23.4 (23.1)	13.1 (13.5)	4.9 (4.5)	77.1 (80.3)	1.0 (1.1)	2.4 (2.2)
57793	27°48'	114°23'	128.2	22.8 (22.6)	14.5 (14.8)	4.1 (4.0)	78.9 (80.8)	1.8 (1.8)	2.3 (2.2)
57799	27°07'	114°55'	78.0	23.9 (23.6)	15.9 (16.0)	4.4 (4.0)	78.6 (78.4)	1.6 (1.5)	2.4 (2.4)
57894	26°35'	114°10'	838.1	23.5 (23.1)	14.7 (14.6)	4.7 (4.5)	74.8 (75.6)	2.0 (2.0)	2.6 (2.5)
57896	26°20'	114°30'	125.3	24.2 (24.0)	15.7 (15.9)	4.6 (4.4)	75.8 (77.3)	1.6 (1.7)	2.6 (2.6)
57993	25°51'	115°00'	124.7	24.9 (24.8)	16.4 (16.8)	4.9 (4.8)	71.2 (74.5)	1.4 (1.5)	2.8 (2.7)
58506	29°35'	115°59'	1165.3	16.3 (16.6)	9.6 (9.6)	4.7 (4.3)	75.7 (78.3)	3.6 (3.5)	2.2 (2.0)
58519	29°00'	116°41'	40.8	22.7 (22.4)	15.5 (15.4)	4.9 (4.8)	73.3 (74.7)	2.1 (1.9)	2.7 (2.5)
58527	29°18'	117°12'	62.9	23.6 (23.6)	14.8 (15.0)	4.8 (4.6)	73.5 (74.2)	1.3 (1.1)	2.5 (2.4)
58606	28°36'	115°55'	45.7	22.6 (22.5)	15.6 (15.7)	5.4 (4.9)	72.3 (73.0)	1.8 (1.8)	2.7 (2.6)
58608	28°04'	115°33'	30.4	23.3 (23.1)	15.7 (15.6)	4.6 (4.2)	75.3 (73.3)	1.2 (1.2)	2.4 (2.4)
58626	28°18'	117°13'	51.7	23.9 (23.3)	15.8 (15.3)	4.6 (4.4)	74.5 (76.1)	1.4 (2.1)	2.5 (2.5)
58634	28°41'	118°15'	113.7	20.0 (19.6)	11.7 (11.8)	3.6 (3.4)	86.4 (84.2)	1.1 (1.3)	1.9 (1.8)
58715	27°35'	116°39'	81.9	23.4 (23.2)	15.1 (15.4)	4.7 (4.6)	78.6 (76.0)	2.6 (2.5)	2.6 (2.7)
58813	26°51'	116°20'	141.6	24.4 (24.2)	15.2 (15.3)	4.5 (4.2)	80.3 (79.1)	1.3 (1.4)	2.4 (2.3)
59102	24°57'	115°39'	302.7	25.1 (24.9)	15.4 (15.8)	4.6(4.4)	77.6 (79.8)	1.2 (1.0)	2.4 (2.3)
Average				**22.9 (22.6)**	**14.6 (14.7)**	**4.6 (4.4)**	**76.4 (77. 1)**	**1.7 (1.7)**	**2.5 (2.4)**

Table 2
Summary of the eight input combinations of meteorological variables used in the three soft computing models.

Combination no.	Model			Input combination
	M5Tree	RF	LightGBM	
1	M5Tree1	RF1	LightGBM1	T_{max}, T_{min}, R_a
2	M5Tree2	RF2	LightGBM2	T_{max}, T_{min}, H_r
3	M5Tree3	RF3	LightGBM3	T_{max}, T_{min}, U_2
4	M5Tree4	RF4	LightGBM4	T_{max}, T_{min}, R_s
5	M5Tree5	RF5	LightGBM5	T_{max}, T_{min}, H_r, U_2
6	M5Tree6	RF6	LightGBM6	T_{max}, T_{min}, H_r, R_s
7	M5Tree7	RF7	LightGBM7	T_{max}, T_{min}, U_2, R_s
8	M5Tree8	RF8	LightGBM8	T_{max}, T_{min}, U_2, R_s, H_r

$$ETo = \frac{0.408(R_n - G) + \gamma\frac{900}{T_a + 273}U_2(e_s - e_a)}{\Delta + \gamma(1 + 0.34U_2)} \quad (1)$$

where R_n (MJ m^{-2} d^{-1}) is the net radiation at the crop surface, which is usually calculated by R_s; G is the soil heat flux density (MJ m^{-2} d^{-1}); T_a is the mean daily air temperature (°C); U_2 is the wind speed at 2 m height (m s^{-1}); e_s and e_a are the saturation and actual vapor pressures (kPa), respectively; $\Delta$is the slope of vapor pressure curve (kPa °C^{-1}) and γ is the air psychrometric constant (kPa °C^{-1}). G can be neglected at a daily time step as in this study. Details of the FAO-56 PM equation can be found in Allen et al. (Allen et al., 1998).

2.3. Simplified empirical models for estimating reference evapotranspiration

Four simplified empirical models, i.e. Hargreaves-Samani, Makkink, Tabari and Trabert, were used for estimating ETo to compare the performances of the three soft computing models. These models were chosen based on the number of required meteorological inputs and their wide applicability to varying climates and regions.

The idea of Hargreaves-Samani model came from the Christiansen (Christiansen, 1968) equation, and the most popular version was Hargreaves-Samani (Hargreaves and Samani, 1985):

$$ET_{o_HS} = aRa\Delta T^b(T_{mean} + c) \quad (2)$$

where *a*, *b* and *c* are the fitted coefficients.

The Makkink model (Makkink, 1957) for estimating ETo using only R_s and T_{mean} data can be expressed as:

$$ET_{o_Mak} = a\frac{\Delta}{\Delta + \gamma}\frac{Rs}{\lambda} - b \quad (3)$$

where *a* and *b* are the fitted coefficients.

The Tabari model (Tabari et al., 2012) is a linear regression model and can be written as:

$$ET_{o_Tab} = aR_s + bT_{max} + cT_{min} + d \quad (4)$$

where *a*, *b*, *c* and *d* are the fitted coefficients.

As mentioned before, mass transfer-based models utilize the concept of eddy transfer of water vapor from an evaporating surface to the atmosphere. The Trabert model (Trabert, 1896) is one of the commonly used model forms:

$$ET_{o_Tra} = aU_2(e_s - e_a) \quad (5)$$

where *a* is the fitted coefficient.

2.4. Tree-based soft computing models for estimating reference evapotranspiration

2.4.1. M5 Model Tree (M5Tree)

Model tree is a kind of generalized regression tree which can extract information and rules from training data (Pourzangbar et al., 2017). The method is to divide the input domain into several sub-domains and establish a linear multivariate regression model (solution) for each subdomain. The final solution is the set of these subdomain solutions. Model trees have multiple subclasses based on different segmentation strategies. M5Tree is one of the most commonly used tree-based models, which takes Standard Deviation Reduction (SDR) as the segmentation criterion. To split the input dataset and generate the final solution, M5Tree takes a top-down approach, using the if-then rules. Firstly, the input data point is entered into the root node. According to the rule of segmentation criterion, the tree is divided into several branches. The process of selecting split nodes and generating branches continues until all data are sorted. In the form of linear multivariate regression, the solution of branch (subdomain) is then calculated on the leaf node of branch. Finally, pruning is essential which can helps to reduce overfitting for branches where the SDR is truly small. More information about M5Tree can be found in Quinlan (1992).

2.4.2. Random Forests (RF)

Random Forest is an integrated soft computing model based on many classification and regression tree (CART) developed by Breiman, 2001. The RF model has been proven to have high predictive accuracy and high tolerance for outliers and "noise". It can also evaluate the relationship between covariables and dependent variables, calculate the relative importance of covariables, and in turn, can be used for feature selecting. The RF model generates *n* data sets with duplicate samples from the original data set. For each dataset, it establishes an unpruned

classification or regression tree. For the regression problem, the final result is the average of all the tree results. However, for the classification problem, the final classification result is selected by all trees voting. Detailed information on the RF model is given by Breiman, 2001.

2.4.3. Light Gradient Boosting Machine (LightGBM)

LightGBM is a gradient learning framework based on decision tree and the idea of boosting (Ke et al., 2017), which is introduced in detail here since it is a relatively new model. Its main difference from the XGBoost model is that it uses histogram-based algorithms to speed up the training process, reduce memory consumption and employ a leaf-wise growth strategy with depth constraints. The basic idea of histogram algorithm is to discretize continuous floating point eigenvalues into k bins and construct a histogram with a width of *k*. Histogram algorithm does not need extra storage of pre-sorted results, and also can save the value after the discretization of features, which is generally enough to store with 8-bit integer, and the memory consumption can be reduced to 1/8 of the original. This rough partitioning does not reduce the accuracy of the model.

Since the decision tree is a weak study model, it is not important whether the segmentation point is accurate or not. The coarser segmentation points also have regularization effect, which can effectively prevent overfitting. The growth strategy of a decision tree is called level-wise, which is an inefficient algorithm because it treats the leaves of the same layer, resulting in a large amount of unnecessary memory consumption. Leaf-wise is a more effective strategy, which finds the leaves with the highest branching gain each time from all the leaves, and then goes through the branching cycle. Therefore, compared with the horizontal direction, the blade can reduce more errors and obtain better precision with the same number of times of segmentation. The downside of leaf orientation is that it can grow deeper decision trees and produce overfitting. Therefore, LightGBM adds a maximum depth limit to the top of the leaf to prevent overfitting while ensuring high efficiency. The schematic diagram of level-wise and leaf-wise tree growth strategies is shown in Fig. 2.

The detailed calculation procedures of LightGBM are described as follows (Sun et al., 2018):

For the given training dataset$X = \{(x_i, y_i)\}_{i=1}^{m}$, LightGBM is aimed to search an approximation $\hat{f}(x)$ to the function $f^*(x)$ for minimizing expected values of specific loss functions$L(y, f(x))$:

$$\hat{f}(x)\arg\min_{f} E_{y,X} L(y, f(x)) \tag{6}$$

LightGBM will integrate many T regression trees $\sum_{t=1}^{T} f_t(X)$ for approximating the eventual model, defined as:

$$f_T(X) = \sum_{t=1}^{T} f_t(X) \tag{7}$$

The regression trees are defined as$w_{q(x)}$, $q \in \{1, 2, ..., N\}$, where N represents the number of tree leaves, q is the decision rule of trees and w is a vector denoting the sample weights of leaf nodes. The model is then trained in the additive form at step t as follows:

$$\Gamma_t \cong \sum_{j=1}^{N} L(y_i, F_{t-1}(x_i) + f_t(x_i)) \tag{8}$$

The objective function is fast approximated by using the Newton's approach. Eq. (8) is simplified when the constant term is removed:

$$\Gamma_t \cong \sum_{j=1}^{N} (g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)) \tag{9}$$

where g_i and h_i are 1st- and 2nd-order gradient statistical results of loss functions. Eq. (9) is further converted to Eq. (10) if the sample set of leaf j is represented by I_j:

$$\Gamma_t = \sum_{j=1}^{J} ((\sum_{i\in I_j} g_i)\omega_j + \frac{1}{2}(\sum_{i\in I_j} h_i + \lambda)\omega_j^2) \tag{10}$$

In terms of the tree structure q(x), the optimum leaf weights of the leaf nodes ω_j^* and extreme values of Γ_K are figured out by Eqs. (11) and (12):

$$\omega_j^* = -\frac{\sum_{i\in I_j} g_i}{\sum_{i\in I_j} h_i + \lambda} \tag{11}$$

$$\Gamma_T^* = -\frac{1}{2}\sum_{j=1}^{J} \frac{\left(\sum_{i\in I_j} g_i\right)^2}{\sum_{i\in I_j} h_i + \lambda} \tag{12}$$

where is the weight function measuring the quality of tree structure q (x). The objective function is eventually obtained by integrating the split:

$$G = \frac{1}{2}\left(\frac{\left(\sum_{i\in I_l} g_i\right)^2}{\sum_{i\in I_l} h_i + \lambda} + \frac{(\sum_{i\in I_r} g_i)^2}{\sum_{i\in I_r} h_i + \lambda} + \frac{(\sum_{i\in I} g_i)^2}{\sum_{i\in I} h_i + \lambda}\right) \tag{13}$$

where I_l and I_r are samples of the left and right branch, respectively.

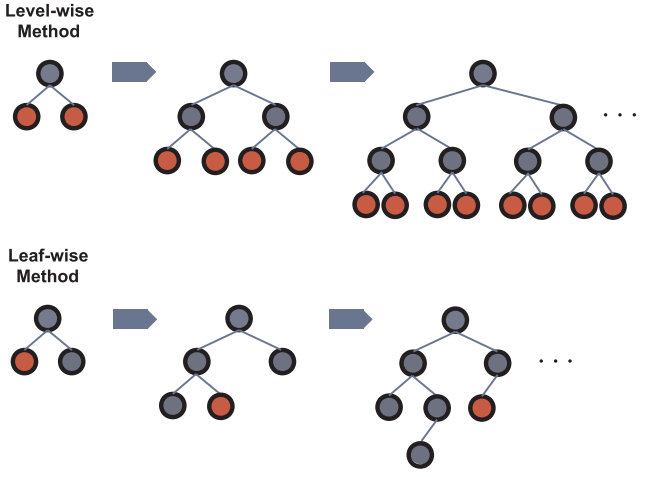


Fig. 2. Schematic diagram of the level-wise and leaf-wise tree growth strategies.

Table 3
Statistical values averaged over the 16 stations for the four empirical models during calibration and validation.

Model	Calibration			Validation		
	R^2	RMSE (mm d^{-1})	NRMSE	R^2	RMSE (mm d^{-1})	NRMSE
Tabari	0.94	0.38	0.16	0.94	0.39	0.16
Makkink	0.92	0.43	0.18	0.93	0.40	0.17
Hargreaves-Samani	0.79	0.70	0.29	0.77	0.72	0.31
Trabert	0.41	1.19	0.49	0.37	1.24	0.52

2.5. Model scenarios and parameter optimization

In local applications, the temperature-based (Hargreaves-Samani), radiation-based (Tabari and Makkink) and mass transfer-based (Trabert) empirical models were used to estimate daily ETo at each station of the 16 stations in Jiangxi Province. These empirical models were calibrated by the FAO-56 PM ETo as reference. LightGBM, RF and M5Tree were then trained and tested using meteorological data from the 16 target stations, which were further compared with the four empirical models. To evaluate the performance of empirical and soft computing models in external (cross-station) applications, ETo at each of the 16 target stations was estimated by models developed with data

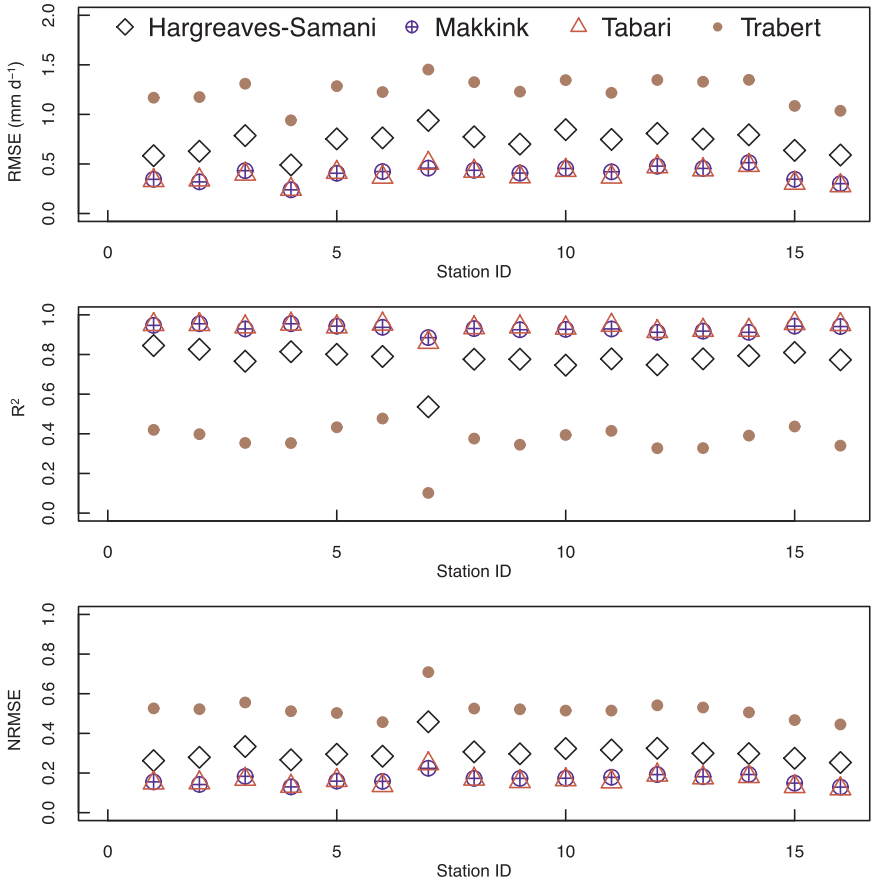


Fig. 3. Statistical indices of the four empirical models at the 16 weather stations during testing.

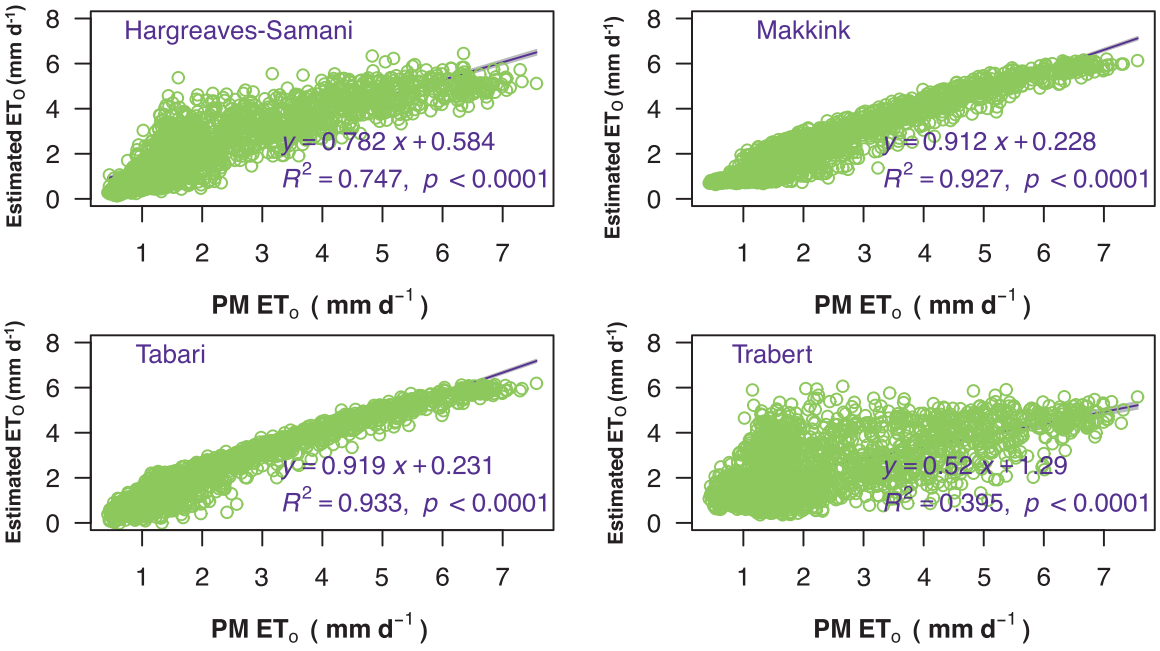


Fig. 4. Scatter plots of ETo values estimated by the four empirical models against FAO-56 PM ETo values at Station 58606 during testing.

from the other 15 stations. In addition, Station 57894 was taken as an example by assuming that this station lacked previous meteorological data. Given that input combinations 1 and 4 are the most common combinations for daily ETo estimation, the soft computing models developed based on these two combinations from the other 15 stations were used to predict daily ETo at Station 57894, which had the most similar climatic characteristics to the other stations. Finally, to further evaluate the performance of the proposed models in larger areas with similar climates, the soft computing models developed with meteorological data from Station 57894 under the input combinations 1 and 4 were used to estimate daily ETo at other 33 weather stations across the humid subtropical region of China as shown in Fig. 1.

Table 4
Statistical values averaged over the 16 stations for the three soft computing models under various input combinations during training and testing.

Model	Input combination	Training			Testing		
		R^2	RMSE (mm d^{-1})	NRMSE	R^2	RMSE (mm d^{-1})	NRMSE
LightGBM1	T_{max}, T_{min}, R_a	0.89	0.50	0.21	0.85	0.58	0.24
LightGBM2	T_{max}, T_{min}, H_r	0.89	0.51	0.21	0.85	0.57	0.24
LightGBM3	T_{max}, T_{min}, U_2	0.92	0.44	0.18	0.90	0.49	0.21
LightGBM4	T_{max}, T_{min}, R_s	0.98	0.23	0.09	0.97	0.27	0.11
LightGBM5	T_{max}, T_{min}, H_r, U_2	0.93	0.39	0.16	0.91	0.45	0.19
LightGBM6	T_{max}, T_{min}, H_r, R_s	0.99	0.13	0.05	0.99	0.18	0.07
LightGBM7	T_{max}, T_{min}, U_2, R_s	0.99	0.16	0.07	0.98	0.22	0.09
LightGBM8	**T_{max}, T_{min}, U_2, R_s, H_r**	**1.00**	**0.03**	**0.01**	**1.00**	**0.08**	**0.03**
M5Tree1	T_{max}, T_{min}, R_a	0.88	0.51	0.21	0.83	0.62	0.26
M5Tree2	T_{max}, T_{min}, H_r	0.88	0.52	0.21	0.84	0.61	0.26
M5Tree3	T_{max}, T_{min}, U_2	0.91	0.45	0.18	0.88	0.53	0.22
M5Tree4	T_{max}, T_{min}, R_s	0.97	0.25	0.10	0.96	0.29	0.12
M5Tree5	T_{max}, T_{min}, H_r, U_2	0.92	0.42	0.17	0.89	0.50	0.21
M5Tree6	T_{max}, T_{min}, H_r, R_s	0.99	0.17	0.07	0.99	0.18	0.08
M5Tree7	T_{max}, T_{min}, U_2, R_s	0.98	0.18	0.08	0.98	0.22	0.09
M5Tree8	**T_{max}, T_{min}, U_2, R_s, H_r**	**1.00**	**0.09**	**0.04**	**1.00**	**0.11**	**0.05**
RF1	T_{max}, T_{min}, R_a	0.97	0.27	0.11	0.84	0.60	0.25
RF2	T_{max}, T_{min}, H_r	0.97	0.28	0.12	0.84	0.60	0.25
RF3	T_{max}, T_{min}, U_2	0.97	0.24	0.10	0.89	0.51	0.22
RF4	T_{max}, T_{min}, R_s	0.99	0.13	0.05	0.97	0.28	0.12
RF5	T_{max}, T_{min}, H_r, U_2	0.98	0.21	0.09	0.90	0.47	0.20
RF6	T_{max}, T_{min}, H_r, R_s	1.00	0.08	0.03	0.98	0.19	0.08
RF7	T_{max}, T_{min}, U_2, R_s	1.00	0.09	0.04	0.98	0.23	0.10
RF8	**T_{max}, T_{min}, U_2, R_s, H_r**	**1.00**	**0.06**	**0.02**	**0.99**	**0.13**	**0.06**

In this study, The RF model was implemented in the R package named "randomForest", while the M5Tree model was implemented in the Matlab package named "M5PrimeLab". The LightGBM package was developed by Microsoft, which can be downed at: https://github.com/microsoft/LightGBM. The coefficients of the four empirical models were attained by the least-squares approach, while the main parameters of the three soft computing models were optimized with the grid search method. For M5Tree, the model parameters of minimum leaf size (MLS) and minimum parents size (MPS) were optimized, with MLS ranging from 5 to 25 at 5 intervals and MPS as 2 times of MLS. For RF, the number of trees (NT) and maximum depth of the tree (MDT) were optimized with NT ranging from 100 to 1000 at 100 intervals and MTD ranging from 3 to 15 at 3 intervals. For LightGBM, NT and MDT were optimized with NT ranging from 100 to 500 at 100 intervals and MTD ranging from 3 to 15 at 3 intervals. The learning was set to 0.05. Basically, all combinations of parameters were tried and the one with the best accuracy was chosen as the optimal parameter combination.

2.6. Statistical indices

Three commonly used statistical indices were used to assess the prediction accuracy of models (Fan et al., 2018a, 2019b; Wu et al., 2019b; Ma et al., 2019), including the coefficient of determination (R^2), root mean square error (RMSE) and normalized root mean square error (NRMSE), defined as follow:

$$R^2 = \left[\frac{\sum_{i=1}^{m} (ET_{io} - ET_{o_mean})(ET_{ie} - ET_{e_mean})}{\sqrt{\sum_{i=1}^{m} (ET_{io} - ET_{o_mean})^2 \sum_{i=1}^{m} (ET_{ie} - ET_{e_mean})^2}} \right]^2 \quad (14)$$

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^{m} (ET_{ie} - ET_{io})} \quad (15)$$

$$NRMSE = \frac{RMSE}{ET_{o_mean}} \quad (16)$$

where ET_{io} and ET_{ie} are the ETo value at i-th time step estimated by FAO-56 PM equation and the other model, respectively; ET_{o_mean} and ET_{e_mean} are the mean ETo values estimated by FAO-56 PM equation and the other model, respectively. Greater R^2 means high prediction accuracy, while smaller RMSE and NRMSE indicate better model performances.

3. Results and discussion

3.1. Performances of empirical and soft computing models in local (target-station) applications

3.1.1. Local application of empirical models

Table 3 represents mean values of R^2, RMSE and NRMSE of the four empirical models averaged over the 16 stations in Jiangxi Province during training and testing. Overall, the statistical results of empirical models showed small differences between the training and testing stages. In the testing stage, the Tabari model yielded the highest R^2 (0.94) but lowest RMSE (0.39 mm d^{-1}) and NRMSE (0.16), followed by the Makkink (R^2 = 0.93, RMSE = 0.40 mm d^{-1} and NRSME = 0.17) and Hargreaves-Samani (R^2 = 0.77, RMSE = 0.72 mm d^{-1} and NRSME = 0.31) models. The Trabert model (R^2 = 0.37, RMSE = 1.24 mm d^{-1} and NRSME = 0.52) performed worst and it was thus not recommend for estimation of ETo in this region.

R^2, RMSE and NRMSE values of the four empirical models for daily ETo estimation at each of the 16 stations during testing are given in Fig. 3. It can be seen from the figure that the statistical results fluctuated among the studied stations, but following the same trends. The Trabert model performed worst at all 16 stations, followed by the Hargreaves-Samani model. The Tabari model was slightly better than the Makkink model at most stations. The highest RMSE value was obtained at Station 58506 in most cases, which can be largely attributed to the low data quality caused by the large variations of meteorological variables as a result of much higher elevation at Station 58506 (1165.3 m) compared with the other stations (most < 150.0 m). According to RMSE values, Station 57894 showed the minimal errors among the empirical models. However, Station 59102 achieved the least NRMSE. The difference between RMSE and NRMSE results can be attributed to the different characteristics of the indices, where NRMSE considered the reference mean value (FAO-56 PM ETo in this study) on the basis of RMSE, while stations with larger ETo values tended to

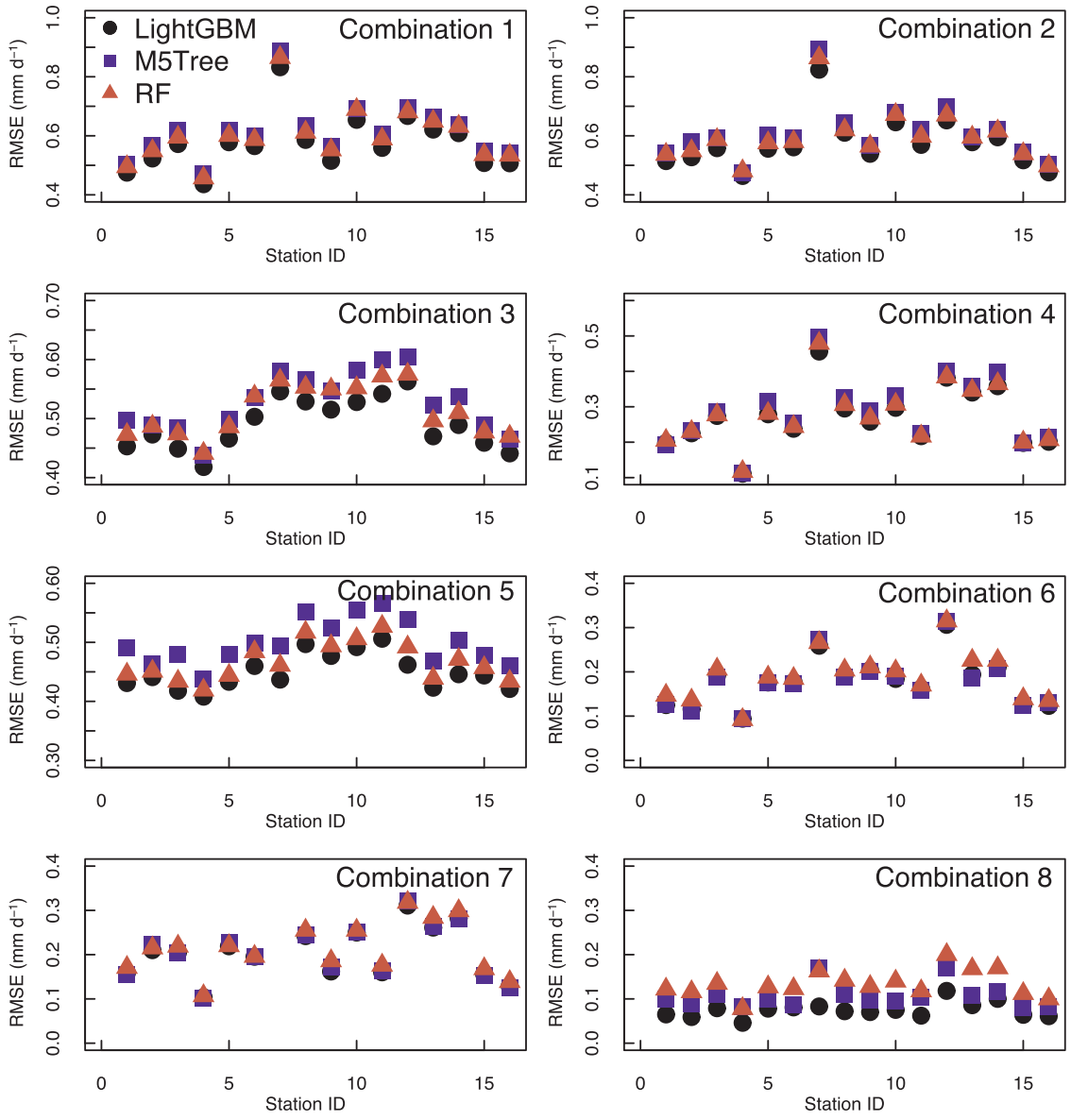


Fig. 5. Statistical indices of the three soft computing models under various input combinations at the 16 stations during testing.

achieve higher RMSE.

Fig. 4 represents the scatter plots between ETo values obtained by the FAO-56 PM equation and four empirical models at Station 58606 (Nachang, as the provincial capital of Jiangxi Province) during testing as an example. The scatter of Tabari model was more evenly distributed between 0 and 8.0 mm d^{-1}, and underestimated ETo when values were less than 3.0 mm d^{-1} and greater than 7.0 mm d^{-1}. The Makkink model showed more scattered points than the Tabari model, and had slight underestimation. The Hargreaves-Samani model significantly overestimated ETo when values were less than 2.0 mm d^{-1}, and mildly underestimated ETo when values were greater than 6.5 mm d^{-1}. The scatter of the Trabert model showed cluster, with only a few scattered points around the trend line. Therefore, it can be concluded that the radiation-based models (Tabari and Makkink) models present the most accurate results among the four empirical models. High performances of radiation-based models have been also reported by Tabari et al. (2016), Chatzithomas and Alexandris (2015) and Feng et al. (2017). In addition, since the Hargreaves-Samani model only requires temperature data, acceptable results can be still obtained in areas lacking complete meteorological elements.

3.1.2. Local application of soft computing models

Table 4 presents mean values of R^2, RMSE and NRMSE averaged over the 16 stations for the three soft computing models under the eight input combinations during training and testing. R^2, RMSE and NRMSE values of the three soft computing models at each of the 16 stations during testing is given in Fig. 5. R^2, RMSE and NRMSE values of LightGBM ranged 0.85–1.00, 0.08-0.58 mm d^{-1} and 0.03-0.24 during testing, respectively. The corresponding values were 0.83–1.00, 0.1-0.62 mm d^{-1} and 0.05-0.26 for M5Tree, while they varied 0.84-0.99, 0.13-0.60 mm d^{-1} and 0.06-0.25 for RF, respectively. It can been seen from Tables 4 and Fig. 5 that LightGBM exhibited lower statistical errors than RF and M5Treeduring testing, while RF showed better prediction accuracy during training. LightGBM was superior to M5Tree and RF under all input combinations. RMSE values of M5Tree and RF were decreased by 0–10.9% and 2.6%–7.3% compared with those of LightGBM in the testing stage, respectively, while the corresponding NRMSE values were decreased by 1.1%–11.6% and 2.7%–8.1%, respectively. The prediction accuracy of M5Tree was worse than that of RF under the input combinations 1–5, while it showed slightly better performance than RF under the input combinations 6–8 at most stations (Fig. 5).

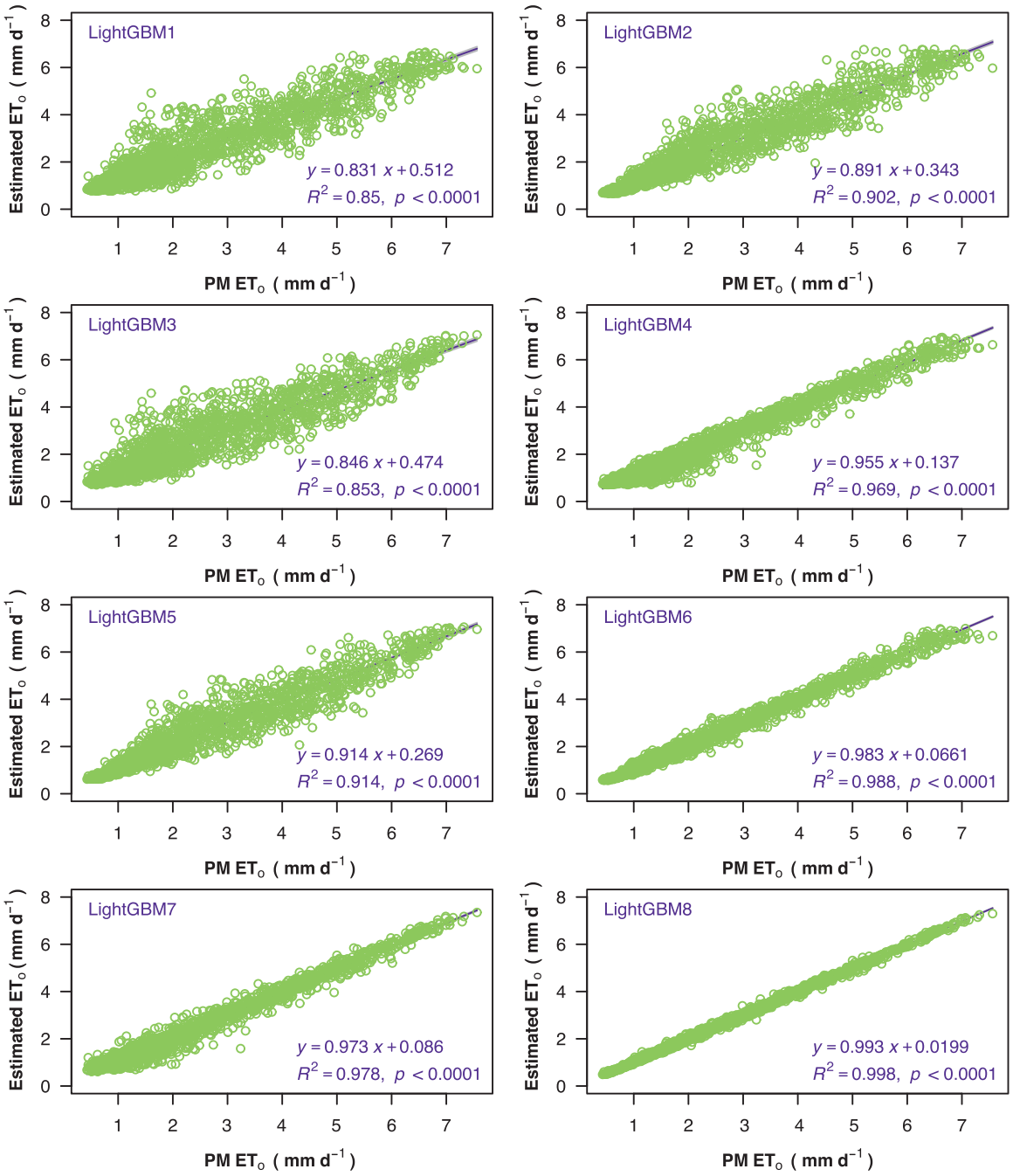


Fig. 6. Scatter plots of ETo values estimated by LightGBM against FAO-56 PM ETo values at Station 58606 during testing.

R_a is a geographical parameter related only to latitude and day of year. It has been found that adding this information to the soft computing models can improve the prediction accuracy of models (Feng et al., 2017). In this study, the input combination 1 (T_{max}, T_{min} and R_a) yielded acceptable ETo estimates. Taking LightGBM as an example, R^2, RMSE and NRMSE values were 0.85, 0.58 mm d^{-1} and 0.24, respectively. When replacing R_a with U_2, H_r or R_s under the input combinations 2–4, the models sometimes gave better performance, with decreases in RMSE by 0.2%, 14.7% and 113.0% for LGBM2, LGBM3 and LGBM4 compared with LGBM1. It seems that U_2 and R_a contributed equivalently for estimating daily ETo in this region. Therefore, when only air temperature and wind speed data are available, wind speed can be ignored considering the high costs of wind speed measurements. In addition, R_s was found to be the most crucial meteorological factor for estimating daily ETo in this region, which was similar to the findings by Fan et al. (2018b) in Wuhan and Guangzhou of China and by Feng et al. (2017) in Sichuan of China. However, this was different from the reported results by Mattar (2018) in Egypt, where wind speed was more important information and adding it to temperature-based models significantly decreased RMSE from 10.20 to 0.58 mm d^{-1}. However, adding R_s data only slightly improved the prediction accuracy of ETo in Egypt (Mattar, 2018) and Burkina Faso (Traore and Guven, 2013). These differences were largely attributed to the fact that the dominant meteorological variables for ETo estimation and their contributions to ETo significantly differed in various climatic regions (Feng et al., 2018; Jiang et al., 2019).

The input combinations 5–7 included three meteorological factors and performed differently from each other. The input combination 7 with R_s data achieved ideal results. Taking LGBM6 and LGBM7 as examples, R^2 values were greater than 0.98, RMSE values less than

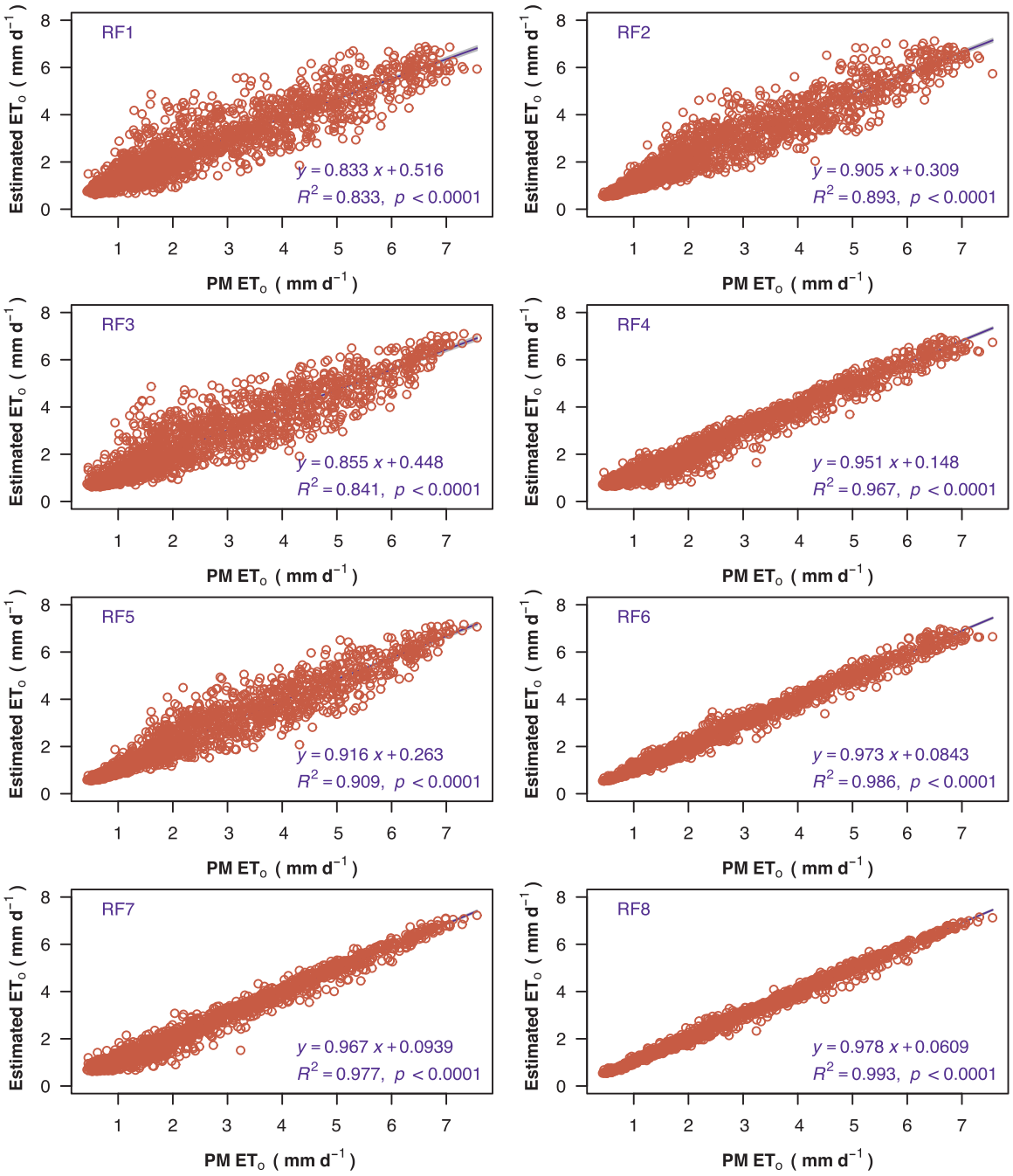


Fig. 7. Scatter plots of ETo values estimated by M5Tree against FAO-56 PM ETo values at Station 58606 during testing.

0.25 mm d^{-1} and NRMSE values less than 0.10. Although LGBM5 did not perform as well as LGBM6 and LGBM7, it did perform better than LGBM2 and LGBM3, with the decrease in RMSE by 8.16% and 21.3%, respectively. There was no significant improvement by adding U_2 or H_r to the input combination 4 (LGBM4), but H_r was found to be slightly better than U_2 for improving the performance of daily ETo estimation. This further indicated that the most important meteorological factor for estimation of daily ETo in this humid subtropical region was R_s, followed by T_{max} and T_{min}, H_r and finally U_2. The input combination 8 had the same input weather factors as the FAO-56 PM formula, except that some intermediate variables were missing, e.g. e_s and e_a. This combination lacked the prospect in practical applications since the reference FAO-56 PM model can be directly applied in this situation, but it can reflect the model capability to simulate nonlinear processes. The results showed that the three soft computing models perfectly simulated the nonlinear process of ETo estimation. Among the models, LightGBM exhibited the best modeling capability. RMSE values were less than 0.10 mm d^{-1} at all 16 stations except Station 58626. Figs. 6–8 present ETo values estimated by the three soft computing models against FAO-56 PM reference values under the eight input combinations. There were clear differences among the three soft computing models under the eight input combinations. The fitting slopes of the input combinations 1–5 were between 0.85 and 0.97, indicating a certain degree of underestimation. In addition, the data points were more scattered than those under the input combination 8. However, input combinations 6-8 had very little underestimation and the data points were closely distributed around the trend line.

3.1.3. Comparison of empirical and soft computing models

The performances of the four empirical models for estimating daily

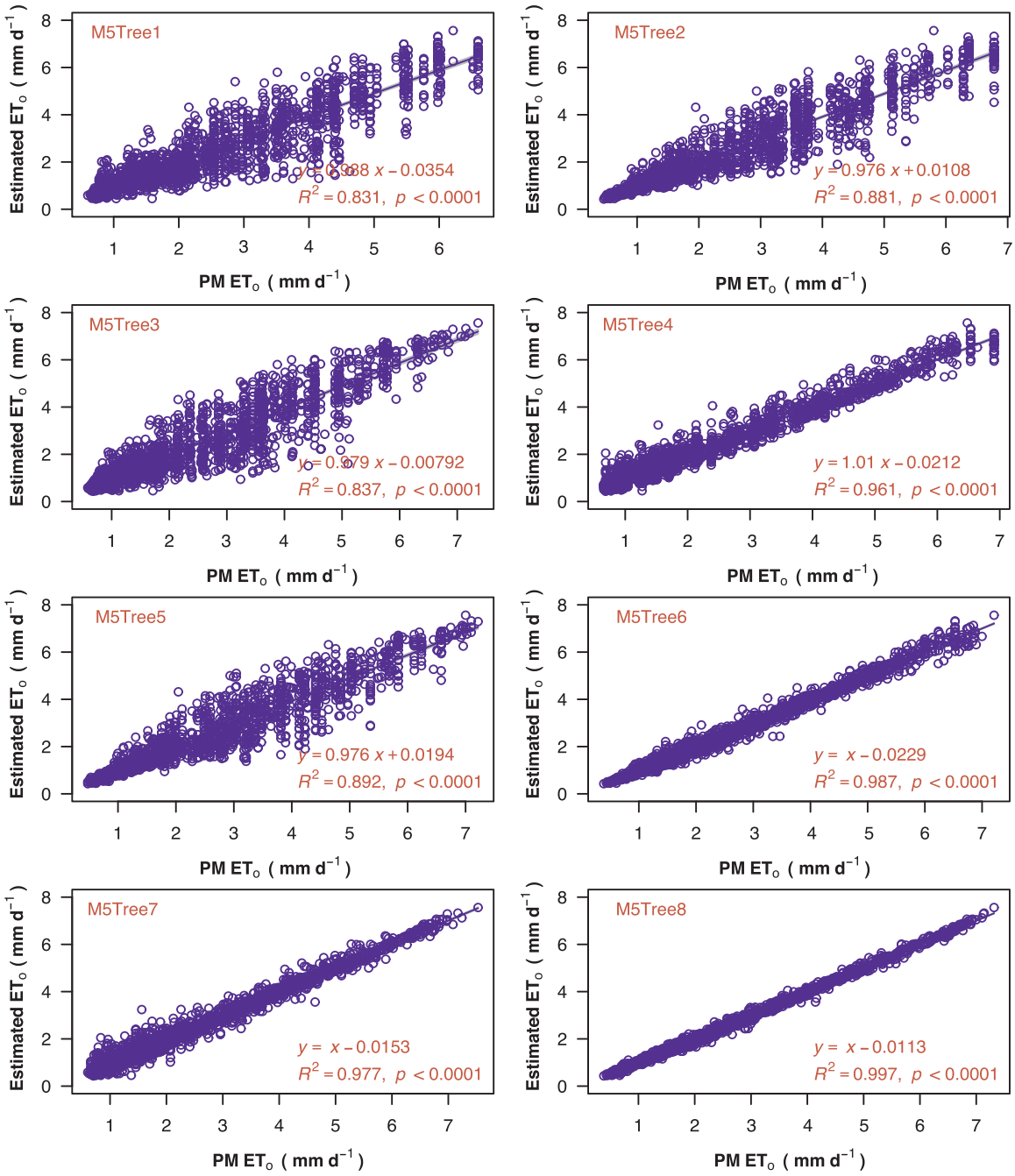


Fig. 8. Scatter plots of ETo values estimated by RF against FAO-56 PM ETo values at Station 58606 during testing.

ETo were further compared with their corresponding soft computing models with the same input variables (Table 5). This section only discussed how each model behaved in the testing period, as the results of these models have been given above. Under the input combination of T_{max}, T_{min} and R_a, LGBM1 was superior to the other models. Compared with LGBM1, RMSE and NRMSE of M5Tree1, RF1 and Hargreaves-Samani models were increased by 7.0%, 4.5%, 25.9% and 6.6%, 4.5% and 25.5%, respectively. Under the input combination of T_{max}, T_{min} and R_s, LGBM4 performed slightly better than RF4, and much better than M5Tree4, Tabari and Makkink models. Compared with LGBM4, RMSE and NRMSE of RF4, M5Tree4, Tabari and Makkink models were increased by 2.6%, 7.4%, 43.0%, 48.9% and 2.7%, 7.1%, 43.4%, 48.7%, respectively. When the input data were T_{max}, T_{min}, H_r and U_2, LGBM5 was the best-performing model. Compared with LGBM5, RMSE and NRMSE of RF5, M5Tree5, Trabert models were increased by 3.8%, 10.9%, 175% and 3.7%, 11.6%, 175%, respectively. Obviously, Trabert model was not suitable for estimating daily ETo in this region as a result of much high statistical errors. However, a comparison of the soft computing and Trabert models indicates that there is great potential to estimate daily ETo based on these information (air temperature, wind speed and relative humidity). How to obtain explicit expression to describe the nonlinear process of ETo based on T_{max}, T_{min} H_r and U_2 needs to be further studied. In general, soft computing models, especially LightGBM, showed much better prediction accuracy than empirical models under the above three input combinations in the studied region.

Table 5
Statistical comparison of empirical and soft computing models with the same input variables during testing.

Model	Input combination	Statistics		
		R^2	RMSE (mm d^{-1})	NRMSE
Hargreaves-Samani	T_{max}, T_{min}, R_a	0.77	0.72	0.31
LightGBM1		0.85	0.58	0.24
M5Tree1		0.83	0.62	0.26
RF1		0.84	0.60	0.25
Makkink	T_{max}, T_{min}, R_s	0.93	0.40	0.17
Tabari		0.94	0.39	0.16
LightGBM4		0.97	0.27	0.11
M5Tree4		0.96	0.29	0.12
RF4		0.97	0.28	0.12
Trabert	T_{max}, T_{min}, H_r, U_2	0.37	1.24	0.52
LightGBM5		0.91	0.45	0.19
M5Tree5		0.89	0.50	0.21
RF5		0.90	0.47	0.20

3.2. Performances of empirical and soft computing models in external (cross-station) applications

3.2.1. External application of empirical models

To evaluate the performance of empirical models for estimation of ETo in external applications, ETo was estimated at each of the 16 target stations using empirical models developed with data from the other 15 stations. The obtained RMSE values of the four empirical models for estimating daily ETo are presented in Tables S2-S5. Similar to local applications, Tabari model produced the lowest average RMSE (0.42 mm d^{-1}), followed by Makkink (0.43 mm d^{-1}) and Hargreaves-Samani (0.85 mm d^{-1}) models. As expected, Trabert model performed worst with very large average RMSE of 1.27 mm d^{-1}. However, those average RMSE values of various empirical models in external applications were quite similar to the corresponding values obtained in local applications (RMSE = 0.39, 0.72, 0.40 and 1.24 mm d^{-1} for Tabari, Hargreaves-Samani, Makkink and Trabert, respectively), indicating that estimation of ETo using empirical models developed with meteorological data observed from cross stations is desirable in this humid subtropical region.

3.2.2. External application of soft computing models

The statistical results of three soft computing models for estimation of ETo at the target station (57894) using models developed with data from the other 15 stations in Jiangxi Province under the input combinations 1 and 4 are presented in Tables 6 and 7. R^2, RMSE and NRMSE values of LightGBM under the input combination 1 ranged 0.72-0.85, 0.44-0.89 mm d^{-1} and 0.24-0.49 during testing, respectively. The corresponding values were 0.52-0.83, 0.47-1.21 mm d^{-1} and 0.26-0.90 for M5Tree, and 0.64-0.83, 0.46-1.06 mm d^{-1} and 0.25-0.58 for RF, respectively. Similar results were also obtained by the three models under the input combination 4. As can be seen from the above results, LightGBM performed significantly better than the other two soft computing models. Compared with local applications, the statistical errors of M5Tree under the input combination 1 in cross-station applications were increased by 44%–154%. The average RMSE value obtained with models developed using data from Station 57799 under the input combination 1 was only 8.9% higher than that obtained in the local application. Besides, developed models with data from Stations 57793, 57896, 57993 and 58527 also yielded no more than 15% increase in RMSE. Under the input combination 4, Station 57598 was the most suitable cross station, with the relative increase in RMSE of 34%. This indicates that the distance between two stations is not necessarily the most important factor for cross-station applications, whereas the similarity of meteorological variables between the two stations actually determines the performance of models with external meteorological data. Of course, the degree of similarity in meteorological variables generally decreases with the increasing distance between two stations (Lu et al., 2018), but it is not always the case.

ETo was further estimated at each of the 16 target stations by soft computing models developed with data from the other 15 stations under the input combinations 1 and 4. The obtained RMSE values of the three soft computing models are presented in Tables S5–S11. As shown in the tables, the average RMSE of LightGBM under the input combinations 1 was 0.68 mm d^{-1} and 0.33 mm d^{-1}, which was 6.7%, 4.5% and 4.9%, 3.2% lower than those of M5Tree and RF, and 20.0% lower than that of Hargreaves-Samani model, respectively. It is clear that LightGBM generally performed better than the other two tree-based models, followed by Hargreaves-Samani model. The lowest RMSE of LightGBM (0.64 mm d^{-1}) and RF (0.68 mm d^{-1}) under the input combination 1 was obtained at Station 57894, which had the most similar climatic characteristics to the other stations. Besides, M5Tree was superior to these models with RMSE of only 0.60 mm d^{-1} based on Station 57894. Under the input combination 4, similar results were achieved, where LightGBM was generally superior to M5Tree and RF,

Table 6
Statistical results of the three soft computing models with T_{max}, T_{min} and R_a as inputs for estimating daily ETo at Station 57894 using models developed with data from the other 15 cross stations.

Cross station	LightGBM			M5Tree			RF		
	R^2	RMSE (mm d^{-1})	NRMSE	R^2	RMSE (mm d^{-1})	NRMSE	R^2	RMSE (mm d^{-1})	NRMSE
57598	0.83	0.51	0.28	0.81	0.68	0.29	0.81	0.52	0.28
57793	0.82	0.47	0.26	0.80	0.68	0.30	0.80	0.50	0.27
57799	0.83	0.47	0.25	0.76	0.79	0.34	0.81	0.49	0.27
57894	**0.85**	**0.44**	**0.24**	**0.82**	**0.47**	**0.26**	**0.83**	**0.46**	**0.25**
57896	0.82	0.48	0.26	0.79	0.79	0.33	0.79	0.52	0.28
57993	0.83	0.48	0.26	0.80	0.76	0.30	0.81	0.51	0.28
58506	0.72	0.89	0.49	0.52	1.21	0.90	0.64	1.06	0.58
58519	0.81	0.65	0.35	0.77	0.87	0.41	0.79	0.71	0.38
58527	0.82	0.49	0.27	0.79	0.69	0.29	0.80	0.51	0.28
58606	0.80	0.72	0.39	0.76	0.96	0.45	0.77	0.76	0.42
58608	0.81	0.50	0.27	0.75	0.79	0.35	0.79	0.54	0.29
58626	0.81	0.50	0.27	0.73	0.85	0.37	0.80	0.52	0.28
58634	0.82	0.51	0.28	0.76	0.81	0.35	0.79	0.54	0.30
58715	0.83	0.52	0.28	0.78	0.88	0.38	0.80	0.56	0.31
58813	0.82	0.54	0.29	0.76	0.75	0.30	0.78	0.57	0.31
59102	0.82	0.54	0.29	0.75	0.70	0.28	0.75	0.59	0.32
Average	**0.82**	**0.54**	**0.30**	**0.76**	**0.79**	**0.37**	**0.78**	**0.59**	**0.32**

Table 7
Statistical results of the three soft computing models with T_{max}, T_{min} and R_s as inputs for estimating daily ETo at Station 57894 using models developed with data from the other 15 cross stations.

Cross station	LightGBM			M5Tree			RF		
	R^2	RMSE (mm d^{-1})	NRMSE	R^2	RMSE (mm d^{-1})	NRMSE	R^2	RMSE (mm d^{-1})	NRMSE
57598	0.98	0.15	0.08	0.98	0.21	0.10	0.98	0.16	0.08
57793	0.98	0.19	0.10	0.98	0.25	0.12	0.98	0.21	0.11
57799	0.98	0.16	0.09	0.96	0.36	0.16	0.98	0.17	0.10
57894	**0.99**	**0.11**	**0.06**	**0.99**	**0.11**	**0.06**	**0.99**	**0.12**	**0.06**
57896	0.98	0.21	0.11	0.97	0.39	0.17	0.98	0.22	0.12
57993	0.98	0.23	0.12	0.97	0.37	0.15	0.98	0.24	0.13
58506	0.96	0.36	0.20	0.86	0.56	0.31	0.95	0.45	0.24
58519	0.98	0.29	0.16	0.96	0.41	0.18	0.97	0.32	0.17
58527	0.98	0.18	0.10	0.96	0.30	0.14	0.98	0.21	0.11
58606	0.98	0.26	0.14	0.96	0.46	0.20	0.97	0.28	0.15
58608	0.98	0.15	0.08	0.97	0.31	0.14	0.98	0.16	0.09
58626	0.98	0.19	0.11	0.94	0.48	0.22	0.98	0.22	0.12
58634	0.98	0.26	0.14	0.95	0.46	0.20	0.98	0.29	0.16
58715	0.98	0.27	0.15	0.95	0.57	0.25	0.97	0.29	0.16
58813	0.98	0.14	0.08	0.98	0.24	0.11	0.98	0.14	0.08
59102	0.98	0.23	0.12	0.97	0.24	0.10	0.97	0.23	0.13
Average	**0.98**	**0.21**	**0.12**	**0.96**	**0.36**	**0.16**	**0.98**	**0.23**	**0.13**

Table 8
RMSE values (mm d^{-1}) of the three soft computing models with T_{max}, T_{min} and R_a as well as T_{max}, T_{min} and R_a as inputs for estimating daily ETo at other 33 weather stations across the humid subtropical region of China using models developed with data from Station 57894.

Cross station	T_{max}, T_{min}, R_a			T_{max}, T_{min}, R_s		
	LightGBM	RF	M5Tree	LightGBM	RF	M5Tree
56196	0.75	0.84	0.85	0.62	0.76	0.74
56651	1.69	1.78	1.78	1.30	1.36	1.40
56666	0.96	1.13	1.06	0.74	0.88	0.93
56739	0.95	1.00	0.99	0.47	0.54	0.57
56778	1.36	1.46	1.45	0.90	0.96	1.00
56959	0.54	0.56	0.51	0.45	0.59	0.57
56985	1.38	1.42	1.42	0.77	0.94	0.95
57178	0.89	0.97	0.96	0.80	0.96	0.95
57245	0.55	0.63	0.61	0.45	0.48	0.49
57461	0.67	0.73	0.71	0.55	0.63	0.61
57494	0.66	0.79	0.78	0.50	0.59	0.60
57516	0.61	0.79	0.76	0.52	0.67	0.70
57649	0.55	0.63	0.60	0.47	0.52	0.52
57687	0.72	0.89	0.85	0.47	0.56	0.60
57816	0.94	1.03	1.01	0.79	0.91	0.91
57874	0.87	1.02	0.99	0.74	0.93	0.93
57957	0.89	0.98	0.99	0.68	0.88	0.86
58141	0.87	0.94	0.94	0.61	0.71	0.70
58208	0.88	0.95	0.93	0.62	0.69	0.68
58238	1.06	1.17	1.16	0.65	0.78	0.79
58265	1.11	1.23	1.22	0.51	0.57	0.57
58321	0.85	0.94	0.94	0.59	0.71	0.71
58362	1.19	1.32	1.29	0.75	0.96	0.96
58457	1.01	1.22	1.17	0.87	1.04	1.05
58531	0.57	0.68	0.65	0.42	0.49	0.50
58665	0.89	1.01	0.99	0.50	0.64	0.63
58737	0.49	0.66	0.62	0.39	0.47	0.48
58847	0.88	1.01	0.99	0.69	0.94	0.93
59287	0.76	0.88	0.88	0.58	0.72	0.70
59316	1.01	1.06	1.06	0.55	0.73	0.72
59431	0.60	0.65	0.64	0.41	0.56	0.52
59644	1.16	1.29	1.34	0.65	0.85	0.83
59758	1.13	1.28	1.31	0.60	0.82	0.79
Average	**0.90**	**1.01**	**0.99**	**0.62**	**0.75**	**0.75**

followed by Tabari and Makkink models, with average RMSE values of 0.30, 0.33, 0.35, 0.42 and 0.43 mm d^{-1}, respectively. Although the most suitable station changed to Station 57793 for LightGBM (0.30 mm d^{-1}) and RF (0.31 mm d^{-1}), M5Tree (0.24 mm d^{-1}) developed using data from Station 57894 still yielded the lowest RMSE. Obviously, the proposed soft computing models, especially LightGBM outperformed their corresponding empirical models under the input combinations 1 and 4. The most appropriate model to represent the whole region did not correspond to LightGBM but M5Tree. However, the developed LightGBM and RF models with data from Stations 57894 also yielded relatively low errors. This may be due to the better meteorological data quality at Station 57894 and its small climatic differences compared with the other 15 stations. Therefore, it was recommended to use soft computing models developed with data from Station 57894 to estimate ETo in the humid subtropical Jiangxi Province, which can greatly simplify the complexity of collecting data from more stations for modeling with ideal prediction accuracy. However, the developed LightGBM, RF and M5Tree models with data from Stations 58506 produced the worst results in all cases, which has been mainly caused by the largest differences in meteorological variables, especially the much lower air temperatures and larger wind speed compared with the other 15 stations (Table 1). This further confirmed that the performance of models with external meteorological data is largely dependent on the similarity of meteorological variables between the cross stations.

To further evaluate the performance of the proposed soft computing models in larger areas with similar climates, LightGBM, RF and M5Tree developed with meteorological data from Station 57894 under the input combinations 1 and 4 were used to estimate daily ETo at other 33 weather stations across the humid subtropical region of China as shown in Fig. 1. As shown in Table 8, average RMSE of LightGBM under the input combinations 1 was 0.90 and 0.62 mm d^{-1}, which was 10.6%, 17.0% and 9.2%, 17.0% lower than those of M5Tree and RF, respectively. This further confirmed that LightGBM was superior to RF and M5Tree for estimation of daily ETo, even applied in much larger regions than where they were developed. However, much larger EMSE values (> 1.1 mm d^{-1}) were obtained at some stations under the input combination 1 with T_{max}, T_{min} and R_a as inputs, such as Stations 56651, 56778, 56985, 58265, 58362, 59964 and 59758 (Table S1). These are largely attributed to the greater differences in meteorological characteristics between Station 57894 and these stations station mostly as a result of longer distances between the stations. However, the performance of soft computing models in external applications was more favorable and stable when more influential meteorological variables were included under the input combination 4 with T_{max}, T_{min} and R_s as inputs. In recent years, it has become increasingly popular to use meteorological data from neighboring stations for estimating ETo at the

target station (Shiri et al., 2015a, 2015b; Lu et al., 2018). This strategy has been approved to work well in various climates around the world, including southwestern China (Feng et al., 2017). The proposed LightGBM model is thus highly recommended to estimate daily ETo in the humid subtropical regions of China and even in other regions around the world with similar climates when lack of local meteorological data. Generally, a long previous dataset are required to develop LightGBM and the other two soft computing models, which may limit the use of this methodology in many areas. However, it is found here that it is still possible to obtain satisfactory ETo estimates at a target station using models developed with data from cross stations with similar meteorological characteristics when previous local data are missing or not available, which can greatly promote the application of the proposed LightGBM model for estimation of daily ETo in data-scare regions.

4. Conclusions

This study assessed the performance of the new tree-based LightGBM model for estimating daily ETo using local and external data from 49 weather stations in the humid subtropical region of China. The tree-based M5Tree and RF along with four empirical models (Hargreaves-Samani, Tabari, Makkink and Trabert) were also compared, based on eight input combinations of daily meteorological data such as maximum temperature (T_{max}), minimum temperature (T_{min}), relative humidity (H_r) and wind speed at 2 m height (U_2) and global solar radiation (R_s) during 2001–2015. The results showed that LightGBM generally outperformed M5Tree and RF in both local and external applications, all of which produced much better daily ETo estimates than the empirical models with the same model inputs. Soft computing models developed with meteorological data from Station 57894 gave satisfactory ETo estimates for the other 15 stations in Jiangxi Province, even for other 33 stations across the humid subtropical region of China. This will provide a new way to simplify the regional model development with meteorological data from as few weather stations as possible in a specific region. LightGBM was thus highly recommended as an efficient alternative model for daily ETo estimation. However, the proposed LightGBM model was only tested in the humid subtropical region of China in the present study, and its applicability in other climates or across various climatic regions remains to be explored. Besides, the parameters of LightGBM were optimized by the traditional grid search method, and the hybridization of LightGBM model with bio-inspired optimization algorithms (e.g. cuckoo search algorithm and flower pollination algorithm) should be explored to further improve its prediction accuracy.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.agwat.2019.105758.

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